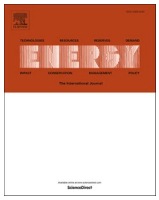
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Field implementation of model-based predictive control in an all-electric school building: Impact of occupancy on energy flexibility

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ABSTRACT

Integrating advanced control strategies is essential in reducing energy cost, optimizing interaction with the electric grid, and enhancing indoor environmental quality in buildings. Enhancing energy flexibility in the building demand profile is essential for the safe and efficient operation of smart grids. This paper presents a grid-interactive model predictive control methodology to improve energy flexibility and maintain indoor environmental quality in school buildings. The proposed model predictive control framework employs data-driven grey-box thermal network models for classrooms with convective heating systems. The methodology is implemented in an all-electric school building in Montreal, Canada, during very cold winter days. Two control scenarios are investigated and compared: 1) a reference scenario using a proportional-integral controller and business as usual thermostat setpoints and 2) a flexible control scenario using model predictive control. Both scenarios were tested under two conditions: with and without occupants in classrooms. Four classrooms operated with proportional-integral controller and usual setpoint profiles were considered reference cases, while another four classrooms employed model predictive control as flexible cases. Results showed that the model predictive control increased energy flexibility by 36 % in unoccupied conditions and 61 % in occupied conditions while reducing energy consumption by 25 % and 63 %, respectively. Throughout both scenarios, the model predictive control maintained satisfactory thermal comfort and indoor air quality. This approach is scalable and transferable to other institutional or mid-size commercial buildings.

1. Introduction

The role of buildings as flexible loads is increasingly recognized as an important component of smart electricity grids; buildings can act as energy generators, energy storage devices, and controllers of demand [1]. Energy flexibility, defined as "the ability to manage demand and generation according to local climate conditions, user needs and grid requirements", is critical to address the grid's challenges of balancing supply and demand and integrating renewable energy sources [2]. In the past, investments have been made by utilities in the distribution system. The resulting cost has spread to consumers, slowing the rate of de-carbonization. New electric technologies allow facing those challenges in a more affordable and predictable way [3]. It is possible today to consider the optimization of the entire electric system from the generation units to the final end-uses. Buildings are part of this system and can bring flexibility to it [4]. To achieve this, research needs to focus on developing emerging technologies and integrating existing concepts within smart energy systems. This includes exploring innovative design and control approaches while leveraging data-driven methods. This adaptability can be achieved through various approaches, including integration of new technologies in smart energy system [5], energy fee rates [6], and advanced data-driven control strategies [7].

In cold climate such as Canada, the morning and evening peaks in winter put a great deal of strain on the electrical grid to provide electric power during extremely cold weather. For example, during winter, the electrical grid of Québec (Canada) is heavily stressed by peak morning

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(6:00 to 9:00) and evening (16:00 to 20:00) loads [8]. This demand-supply mismatch, especially acute during school start times, underscores the need for dynamic load management strategies. Also, this issue is particularly important for all-electric buildings that rely entirely on electricity for space heating. Therefore, it is essential, especially in cold climates, to enhance the energy flexibility of buildings to support peak load reduction, improve grid reliability, and reduce operational costs [9].

In this context, model predictive control (MPC) has demonstrated its potential to reduce building energy consumption, operational costs, and peak demand [10]. MPC can dynamically adjust Heating, ventilation and air-conditioning (HVAC) settings and optimize energy usage by utilizing real-time data and weather forecasts. This results in significant energy savings and flexible load management for grid services, such as load shifting and peak demand management [11]. Compared to major HVAC system retrofits or building envelope upgrades, implementing advanced control strategies like MPC is often more cost-effective and feasible, as it can be deployed with minimal infrastructure modifications. However, despite extensive research, the adoption of MPC in the building sector remains limited, primarily due to the complexity of controller design and implementation [12]. This study presents an MPC framework that acts as a supervisory controller for existing conventional control systems in buildings, optimizing heating demand response while maintaining energy efficiency and occupant comfort.

School buildings account for 30 % of the public sector's energy use [13] and have an average energy intensity of 200 kWh/m^2 per year in Canada [14]. To put this into perspective, this energy use intensity is about 48 % higher than that of the residential sector (135 kWh/m^2 per year). HVAC energy accounts for 46 % of the energy consumption of school buildings [15]. Addressing energy problems in school buildings is an important component of energy efficiency in the building sector [16]. Thus, transitioning to energy-flexible buildings responding to grid and occupants' requirements is important [17]. Clauss et al. [18] implemented schedule-based control strategies for pre-heating a school building in Elverum, Norway. Their results showed a 50 % peak load reduction during on peak hours, while maintaining thermal comfort. However, they highlighted that the lack of automated setpoint adjustments in traditional building management systems (BMS) remains a significant barrier for building managers to adopt energy flexibility measures like pre-heating. This highlights the need for automated approaches, such as MPC, which can optimize energy flexibility measures and improve energy performance effectively.

Cotrufo et al. [19] proposed an MPC framework based on machine-learning techniques for institutional buildings. Their results showed a 22 % reduction in natural gas consumption and a 4 % reduction in the heating demand. Wang et al. [20] compared the performance of reinforcement learning (RL) and MPC in enhancing building energy efficiency and operational flexibility. They found that the MPC controller outperformed the baseline controller, reducing discomfort by 12 % and increasing energy saving by 22 %. Hu et al. [21] simulated an MPC controller for the living room of a residential apartment in Denmark. Their results demonstrated that the MPC controller reduces electricity costs by 19 % compared to conventional on-off controllers. Sun et al. [22] developed a dynamic day-ahead schedule to optimize the load profile and enhance the integration of thermal energy storage in a hybrid energy system. Their results showed that with the dynamic day ahead schedule, the heat pump could operate in the higher COP region, and the energy storage efficiency increased by 3.5 %. Zhao et al. [23] compared the operation of a radiation air conditioning system using MPC with conventional rule-based control and PID control. They showed that the overall energy consumption is reduced by 15 % compared to conventional rule-based control and 8 % compared to the PID [23].

Although numerous studies on the use of MPC in building applications have been conducted and published over the past decades, most of these studies are simulation-based, with only a few papers reporting MPC demonstrations in actual buildings [24]. This indicates the need for more research involving measurements and demonstrations to validate the effectiveness of MPC in real buildings.

2. An Overview of model predictive control field implementation

A few studies implemented MPC in institutional buildings. Ham et al. [25] presented a field demonstration of MPC at a K–12 school in California, US. Their results showed that the MPC strategy could achieve a 24 % reduction in peak demand during the cooling season and up to 16 % during the heating season while allowing end users to override thermostat setpoints. Bruno et al. [26] implemented MPC in a public school equipped with battery and photovoltaic systems in southern Italy. They found that 4 %–7 % electricity cost savings can be achieved compared to commercial rule-based control. Merema et al. [27] implemented an MPC controller for all-air systems in a university located in Ghent, Belgium. Their results showed 10 %–40 % reduction in electric energy use and 21 %–55 % in thermal energy use. Aswani et al. [28] investigated the implementation of an MPC controller for an air conditioner that cools a computer laboratory on the Berkeley campus in California, US. Their results showed that the MPC controller result in a 30 %–70 % reduction in cooling energy consumption compared to the traditional controller.

Moreover, some studies implemented MPC in offices or commercial buildings. Drgoňa et al. [29] developed an MPC framework for an office in Belgium equipped with a ground-source heat pump and thermally activated building structures. Their study found that the MPC approach reduced the building's energy consumption by 54 % and improved thermal comfort by 37 %. Choi et al. [30] presented a low-cost method for implementing MPC using a communication system. A grey-box building and HVAC model were developed for simulations and real-world testing. Their results showed that MPC achieved up to 34 % cost savings compared to a feedback controller. Bengea et al. [31] developed an MPC controller for a multi-zone, variable-volume HVAC system in a mid-size commercial building, which resulted in 75 % reductions in heating costs over the heating season. Kim et al. [32] presented an MPC framework to coordinate multiple rooftop units for medium-sized commercial buildings located in Los Angeles, U.S. They concluded that cooling energy savings of 15 %–30 % can be achieved through applying MPC. Vivian et al. [33] reported the results of MPC performed in a building lab in Piacenza, Italy. They found that this strategy leads to electricity savings of up to 17 % compared to a conventional control. Kun et al. [34] investigated an MPC controller in a convenience store equipped with photovoltaics and batteries in California, USA. The results show that the MPC controller achieved 12 % annual electricity cost savings, and 34 % peak demand reduction compared to a constant setpoint temperature. Table 1 summarizes the results of some studies that have implemented MPC in real buildings from 2005 to 2024.

The literature review highlights that field studies on MPC in school buildings are rare. To the authors' knowledge, no research has implemented MPC in school buildings with a focus on impact of occupancy on MPC performance. The review also reveals that field test studies on energy flexibility—especially in school buildings—are limited. Most investigations remain focused on energy efficiency rather than flexibility. However, with the growing trend toward building electrification and grid-interactive operations, energy flexibility is becoming increasingly important. This underscores the need for more field validation studies to explore this critical aspect. By adopting MPC in school buildings, educational institutions can enhance energy flexibility, improve comfort, and achieve cost savings, ultimately creating a better environment for students and staff.

To address these gaps, this paper presents a grid-interactive MPC solution and field implementation results from an all-electric school during very cold winter conditions, with a specific focus on evaluating

Table 1
Studies on the implementation of MPC in real buildings.

Year	Ref	T_{out}	Building Type	Test duration	Systems controlled	Summary of results
2025	This study	−10 – 5 °C	School classrooms	Three weeks	Heat pumps	36 % power reduction in unoccupied conditions and 61 % in occupied conditions
2024	[30]	22–30 °C	Office	Three days	Fan coils	34 % cost savings compared to a feedback controller
2023	[35]	18–28 °C	School classrooms	One month	RTUs	24 % reduction in total power and 30 % in HVAC power
2023	[11]	26–34 °C	Residential	30 Days	Air conditioner	22–29 % cooling energy saving
2023	[33]	0–5 °C	laboratory	Three days	Fan coils	10–17 % electricity cost saving
2022	[32]	NA	University lab	Two months	RTUs	3.6 %–8.7 % energy cost savings; 36 % demand savings
2022	[34]	NA	Small Commercial	Several months	RTUs, refrigeration, battery	12 % energy cost savings; 34 % demand reductions
2022	[36]	8–22 °C	Office	Two months	Rooftop units, underfloor air distribution system	40 % energy saving
2021	[37]	−2 – 12 °C	Laboratory	32 days	hydronic radiator	22.5 % energy cost saving
2020	[38]	13–39 °C	Residential	Six days	Cooling ceiling panels	24.9 % cooling energy saving
2019	[39]	0–30 °C	Office	20 days	Hydronic radiant floor	50 % cooling, 29 % Heating energy saving
2018	[40]	15–30 °C	Conference room	Four months	Packaged air conditioners	12 % HVAC energy savings and 18 % HVAC peak demand
2016	[41]	2–12 °C	Office	Winter season	Heat pumps, gas boiler	34–40 % energy cost savings; 20–30 % primary energy reduction
2014	[31]	16–38 °C	Laboratory	Several months	Multi-zone HVAC unit	20 % energy savings during the transition season, 70 % during heating season, and 10 % peak power reduction
2011	[42]	3.4 °C	Large building	Winter season	Ceiling radiant heating	Energy saving between 15 % and 28 %
2005	[43]	12–28 °C	Small room	Five days	Chillers and an ice-based TES	17 %–27 % Cost saving

the impact of occupancy on the system's performance. The main contributions of this paper include the following.

- Development of archetype models for zones with convective systems and zones with radiant floor systems in school buildings, which can be used in other mid-size institutional/commercial buildings. The models can capture the dynamics of the indoor temperature variations and the power demand of the HVAC system. The developed models are calibrated with measured data.
- Development of a heuristic MPC methodology to activate energy flexibility in buildings is proposed. The framework offers a practical method for implementing MPC as supervisory control in buildings. The methodology is scalable and can be transferred to other buildings.
- Implementing the developed MPC framework in a school building during the heating season, analyzing its performance for both unoccupied and occupied conditions, and comparing it with business-as-usual cases with a PI controller.

3. Methodology

As illustrated in Fig. 1, four main steps are followed to develop a simulation model that enables the evaluation of different control strategies.

- **Data collection:** Operational data (including indoor air temperature, electricity demand, occupancy levels, and CO2 level of zones with convective heating systems) were collected. Data was collected during the winter season, capturing both occupied days (during normal school operations) and unoccupied periods (during spring break). Weather data was obtained from local weather stations for the time of the experiments and interpolated when necessary.
- **Model development:** While various models can be used to capture a building's thermal response to control actions, grey-box models have demonstrated significant potential in accurately representing the thermal dynamics of buildings. Resistance-capacitance (RC) thermal network models, corresponding to an archetypical configuration

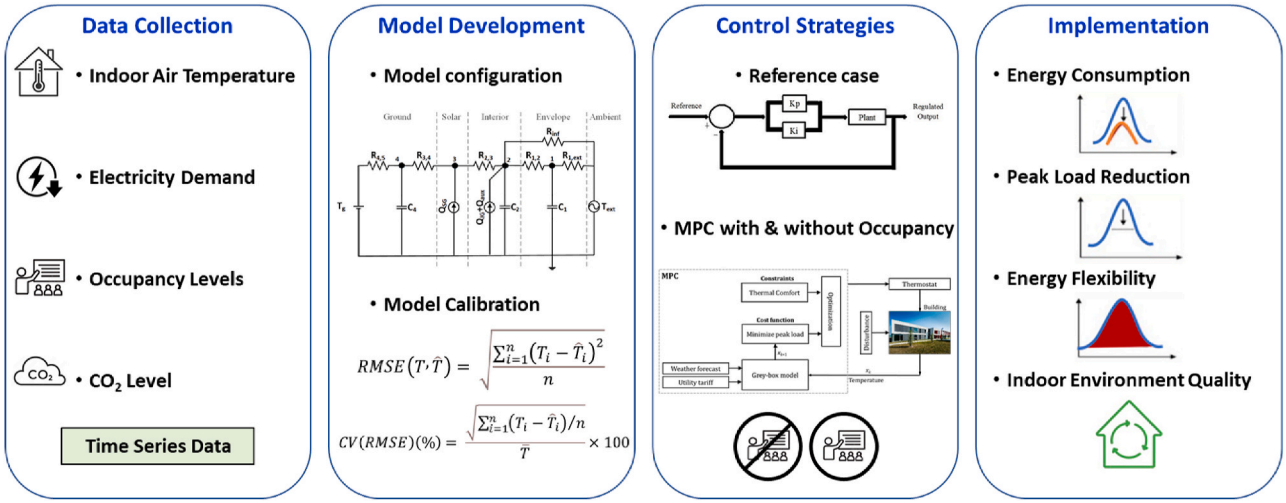


Fig. 1. Schematic of the methodology in this paper.

[42], were developed and calibrated for both occupied and unoccupied scenarios for zones with convective system.

- **Control strategies:** An MPC algorithm was developed to optimize temperature setpoint profiles over the next 24-h period using real-time data and weather forecasts. The objective of the MPC is to reduce peak demand during high electricity price periods, while maintaining thermal comfort and indoor air quality. The MPC framework was applied to two scenarios: 1) without occupancy, implemented during unoccupied periods and 2) with occupancy, applied during normal school hours.
- **Field implementation and energy flexibility assessment**: Four control scenarios were investigated for cold winter days. 1) PI controllers were employed in unoccupied classrooms, serving as a reference for baseline performance without occupancy (Reference I). 2) PI controllers were applied in occupied classrooms, representing the reference case with occupancy (Reference II). These controllers responded to real-time data and changing conditions, serving as the baseline control strategy commonly used in conventional systems. 3) MPC was implemented in unoccupied classrooms (Flexible I). 4) MPC was applied in occupied classrooms, enabling more realistic temperature control and enhancing energy savings during peak morning hours (Flexible II). In these flexible cases, the MPC system optimized energy consumption by adjusting temperature setpoints based on real-time data and weather forecasts. The optimized temperature profiles were applied to the classrooms for the next 24-h period. The results from the flexible MPC scenarios are compared with the reference PI scenarios in terms of energy consumption, peak load reduction, and energy flexibility.

3.1. Data collection

Measurements from a winter season of operational observations were used to characterize the model and perform parameter identification. Initially, the total electrical consumption of the school was divided into two components: (a) influenced by the control algorithm and (b) attributed to uncontrolled device (for example, the plug loads and lighting are driven by occupancy and the school's operation schedule).

The classrooms are heated and cooled through 36 local water-to-air heat pumps using water from geothermal boreholes. The HVAC system is driven by ambient conditions, the operation schedule, and occupancy via CO2-based demand-driven fresh-air ventilation. A system-level coefficient of performance (COP) of 3 for cooling and 3.5 for heating is assumed for the operation of the ground source heat pump (GSHP), circulation pumps, and fans. Electrical consumption can be grouped based on the day of the week since the school operates on a weekly schedule, and similarities between each day were expected. Fig. 2 shows the indoor air temperature and power demand of a classroom with Proportional-Integral (PI) controller during normal operation of the school.

Considering that passive infrared occupancy detection sensors are available in the school, occupancy levels are available. The number of students in each classroom is 20 ± 2. Fig. 3 presents normalized occupancy level of a classroom. Improved occupancy data could enable

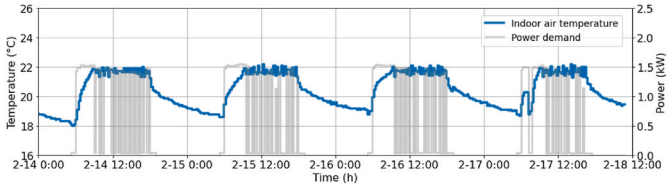


Fig. 2. Indoor air temperature and power demand of a classroom with a PI controller.

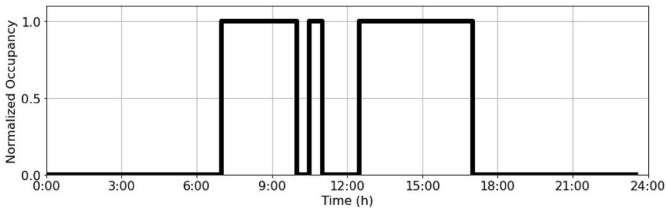


Fig. 3. Normalized occupancy level of a classroom.

enhanced prediction models, shifting the building operation from a fixed schedule to occupant-driven operation, thereby achieving higher energy savings and improving comfort.

3.2. Weather data

Weather data was obtained from the SIMEB (in French, *Simulation Énergetique des Bâtiments*) website [44], which provides a variety of meteorological parameters, such as ambient temperature, humidity, and solar irradiation, from several weather stations in Québec. Data for outdoor temperature and solar irradiance were gathered from Mirabel, the nearest weather station, and linearly interpolated to 15 min. In addition, the MPC application required real-time weather forecasts, which was obtained from Solcast [45].

Fig. 4 presents a sample of weather data collected over a) an unoccupied period and b) an occupied period (normal school operation) during winter. The experiment was conducted on particularly cold days, as electrical power demand typically peaks under such conditions. Fig. 4 illustrates the outdoor temperature and solar irradiance recorded during the study. The outside air temperature drops to −10 °C during the experiment period and the solar irradiation on a horizontal surface with a maximum value above 500 W/m^2 represents sunny days, and 200 W/m2 represents cloudy days.

3.3. Model development

Model development for testing and evaluating control scenarios takes a significant share of the effort required to develop and implement advanced control strategies. Solutions for developing smart supervisory control must be simplified, cost-effective, and easily replicable [46]. It is essential to identify thermal models with sufficient resolution, robust performance, and low computation time [47]. Control-oriented archetypes are simplified energy models that represent generic buildings equipped with generic configurations of building systems and are used to develop, test, and evaluate control decisions. They can be used for decision support and automated supervisory control; they could rely on

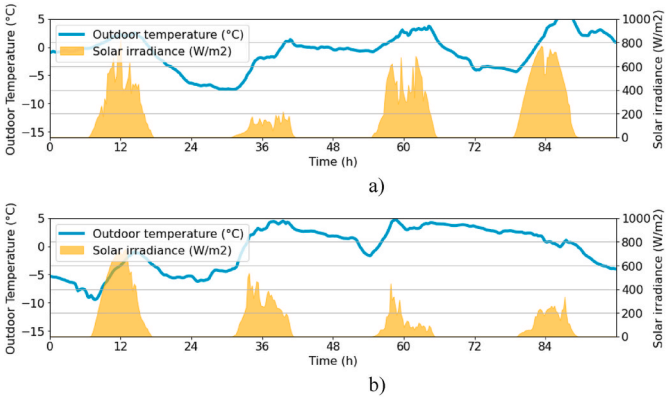


Fig. 4. Montreal weather data. Outdoor air temperature and solar irradiance on a horizontal surface in winter a) during umoccupied period and b) during occupied period.

extensive operational data or limited information. They show a significant potential for replicability by leveraging research on similar buildings and building systems.

The thermal modeling approach adopted in this study follows a lumped-parameter finite difference method, where the thermal behavior of building zones is represented using resistance-capacitance (RC) thermal networks. In this framework, thermal resistances (R) capture the lumped conductive paths between temperature nodes, while thermal capacitances (C) represent the lumped thermal storage properties of zone elements such as air, envelope, and floor layers. Although the governing equations are derived from explicit thermal energy balance principles, the associated parameters are not directly measured but are instead identified through optimization using measured operational data [48]. This modeling strategy thus integrates a physically based model structure with data-driven parameter identification. According to established conventions in building modeling literature [49], such a hybrid approach is referred to as a grey-box model, balancing physical interpretability and empirical accuracy. This characterization is particularly suitable for control-oriented applications, such as model predictive control, where model simplicity and calibration with real data are both essential.

The following steps are used to develop grey-box models and quantify energy flexibility.

- Real building measurement data are collected from the smart meters installed in the classrooms. Data included variables such as power demand (kW), zone air temperature (°C), weather data, and specific data related to each zone (e.g., air temperature). All measurements were taken at intervals of 15 min.
- Numerical models of thermal building control are developed. These models are based on RC thermal network models.
- The developed models are calibrated using the collected data. The parameters are identified using the Sequential Least Squares Programming (*SLSQP*) in Python.
- The objective of the optimization algorithm is to find the equivalent RC circuit parameters by minimizing Coefficient of the Variation of the Root Mean Square Error (CV (RMSE)) and normalized Mean Bias Error (NMBE).
- Appropriate control strategies for zones with the convective system and zones with radiant floor system are presented to enhance the energy flexibility available from the building to the grid at specific times, depending on the grid requirement.
- A Building energy flexibility index (BEFI) is applied to quantify dynamic building energy flexibility at the zone and building levels.

Fig. 5 shows different developed RC thermal networks from first-order to fourth-order models. The zone envelope capacitance is represented by C_1, the air capacitance by C_2, the floor capacitance by C_3, and the second capacitance for floor by C_4. Thermal resistance $R_{1,2}$ describes the resistance between the envelope and the interior air, $R_{2,ext}$ describes the resistance between the envelope and the exterior air, R_{inf} represents the infiltration between the interior air and the outside air, $R_{2,3}$ refers to the resistance between the interior air and the floor, $R_{3,4}$ and $R_{4,5}$ refer to the floor resistances.

A fully explicit finite difference approach is used to solve the energy balance equations at each node in the models. The fully explicit approach assumes that the current temperature of a given node depends only on its temperature and the temperature of the surrounding nodes at a previous time step. By performing a heat balance on the control volume, the differential equation of a node can then be written as [50]:

$$C_i\frac{dT_i}{dt} = Q_i + \sum_{ij}^{n}\frac{(T_j - T_i)}{R_{i,j}} \tag{1}$$

Each node's energy balance is solved using an explicit finite difference method. The discretization of the time derivative term is as follows:

$$C_i\frac{dT_i}{dt} \approx C_i\frac{\Delta T_i}{\Delta t} = C_i\frac{\left(T_i^{t+1} - T_i^{t}\right)}{\Delta t} \tag{2}$$

Temporal discretization of Equation (3) can be rearranged in the following explicit manner. To assure numerical stability in the solution, the time step must be chosen according to the stability criterion

$$T_i^{t+1} = \frac{\Delta t}{C_i}\left[\sum_j U_{ij}^{t}\left(T_j^{t} - T_i^{t}\right) + \sum_j U_{ik}^{t}\left(T_k^{t} - T_i^{t}\right) + \dot{Q}_i^{t}\right] + T_i^{t}$$

$$\Delta t \le \min\left(\frac{C_i}{\sum_{j=1}^{n} U_{ij}}\right), \forall\, i \in \{1, 2, \ldots, n\} \tag{3}$$

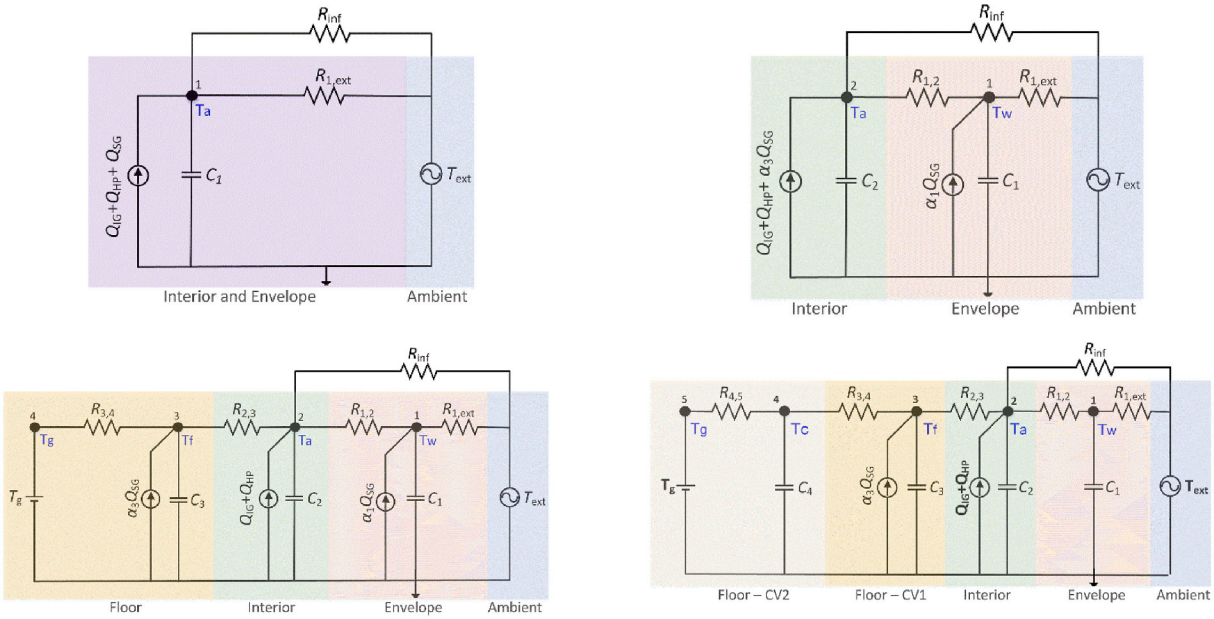


Fig. 5. RC thermal network model of the convectively-heated zones.

Where:

U_{ij}: Conductance between nodes i and j, W/K;

U_{ik}: Conductance between nodes i and k where node k has a defined or known temperature, W/K;

T: is the temperature at node i or adjacent nodes, °C;

C_i: Capacitance of the node $\left(C_i = \rho C_p A dx C_T\right)$, J/ K;

$\dot{Q}$: Heat flow into the node, W;

Δt: Time step, s.

The capacitance of the air node contains a factor C_T (air thermal capacitance multiplier) that accounts for phenomena (furniture and objects, the time required for air mixing, delay due to ducting and other factors) that in a low-order model result in capacitance with an observed effective value significantly larger than the one calculated using only the physical properties of the air. The air thermal capacitance multiplier range between 6 and 10 [51]. Equation (4) in matrix form is:

$$\begin{Bmatrix} T_1 \\ \vdots \\ T_N \end{Bmatrix}^{t+1} = \begin{Bmatrix} \frac{\Delta t}{C_1} \\ \vdots \\ \frac{\Delta t}{C_N} \end{Bmatrix} \odot \left(\begin{bmatrix} -\sum_j^N U_{1j} - \sum_k^M U_{1k} + \frac{C_1}{\Delta t} & U_{12} & \cdots & U_{1N} \\ \vdots & \vdots & \ddots & \vdots \\ U_{N1} & U_{N2} & \cdots & -\sum_j^N U_{Nj} - \sum_k^M U_{Nk} + \frac{C_N}{\Delta t} \end{bmatrix} \begin{Bmatrix} T_1 \\ \vdots \\ T_N \end{Bmatrix}^t + \begin{Bmatrix} \dot{Q}_1 + \sum_k^M (U_{1kk} T_{kk}) \\ \vdots \\ \dot{Q}_N + \sum_k^M (U_{Nkk} T_{kk}) \end{Bmatrix} \right) \tag{4}$$

where.

- $\odot$: is an element-wise multiplication operator.
- N: is the number of nodes.
- M: is the number of nodes with known temperatures.

The objective of the optimization algorithm is to find the equivalent RC circuit parameters by minimizing *CV (RMSE)* and *NMBE*. Equations (5) and (6) [52] are used to calculate this index where T_i , $\widehat{T}_i$, n, and $\overline{T}$ represent the measurements data, simulation results, a total number of observations, and the average of all measurements, respectively.

$$CV(RMSE)(\%) = \frac{\sqrt{\sum_{i=1}^{n} (T_i - \widehat{T}_i) \Big/ n}}{\overline{T}} \times 100 \tag{5}$$

$$NMBE(\%) = \frac{\sum_{i=1}^{n} (T_i - \overline{T})}{(n-1) \times \overline{T}} \times 100 \tag{6}$$

According to ASHRAE Guideline 14 (Measurement of Energy and Demand Savings), the model shall have a CV(RMSE) of not more than 15 % and NMBE of 10 % relative to hourly calibration (measured) data [52].

The simplest first-order model lumps all heat transfer mediums into a single node. This model captures the short-term trends well but would need to be more stable in its parameters. This approach would be suitable for quantifying the heating or cooling the building would require throughout the day and preparing accordingly. A higher-order model is necessary for a finer analysis to represent the building's various components. By putting more importance on the room air, the parameters relating to the slab are changed; no significant effect on the air parameters can be observed. The sum of the capacitances from the second-order model is close to the capacitance from the first-order model. The conductance towards the exterior is also reasonably close, signifying that it can grow the RC network dynamically. The modeling can be further enhanced by separating the floor, envelope, and air nodes, represented by a third or fourth-order model. Results showed that the third-order model represents the minimal number of nodes required to forecast the indoor temperature accurately. Fig. 6 presents the validation results of the measured and estimated temperature of the different classrooms for the third-order RC model. The estimated temperatures capture the trend of the measured temperatures quite well. The CV (RMSE) of the model is 8 ± 3 % and NMBE of 4 ± 2 %, which is acceptable. Table 2 provides a detailed comparison of grey-box model predictions and measured data across different classrooms.

The reduced-order models obtained with the proposed approach are used to be applied in an MPC strategy for the control of the indoor air temperature. The model reduction method is an effective tool to design predictive strategies, in which low-order models are preferable with respect to complex and detailed models. A potential application is the simulation of energy demands in districts where buildings will play an active role in the near future. Thus, by having a control-oriented model of the building, dynamic controls can be used that track the price of electricity and minimize operating costs while guaranteeing the comfort of its occupants.

3.4. Model predictive control

MPC is a control strategy that uses a mathematical model of the system to predict its behavior and optimize its performance. The MPC algorithm generates a control signal that considers the system's current state and predicted behavior over a future time horizon. The MPC controller is trying to minimize the peak load and cost of electricity over a prediction horizon PH, while maintaining indoor environment quality. The optimal actions are implemented for a period called control horizon when the optimization is run again, and the MPC controller updates the optimal actions [53]. This iterative process allows the MPC controller to continuously adapt and optimize the control actions based on real-time data and changing conditions. The general MPC formulation is presented in Equation (7) [10], which includes the objective function, constraints, and the prediction model used in the MPC controller.

$$\min_{u^0,\ldots,u^{N-1}} J \quad where\ J = \sum_{i=1}^{N} l(x^i, u^i, w^i)$$

$$\text{Subject to } x^{i+1} = f(x^i, u^i, w^i) = Ax^i + Bu^i + Ew^i \quad \text{System dynamics}$$

$$h(x^i, u^i, w^i) = 0 \; g(x^i, u^i, w^i) \geq 0 x^1 = x, \text{Current state} \tag{7}$$

Where x is system variables vector, u indicates controllable variables, w is uncontrollable inputs such as weather, l is the cost function, the h function corresponds to equality constraints, and g presents inequality constraints (boundary conditions).

In this study, the objective function aims to minimize electricity costs associated with electricity peak demand while ensuring comfort levels in the building. The constraints consider factors such as power limits, temperature setpoints, and occupancy schedules to ensure feasible and efficient control actions. Equation (2) presents the objective function, where the objective is to minimize the peak load when the energy cost varies over the day, with higher prices during peak hours for the grid. This study concentrates on the electricity rates called *Rate Flex M*, applied in Québec to the medium-sized commercial building sector [54], as presented in Table 1.

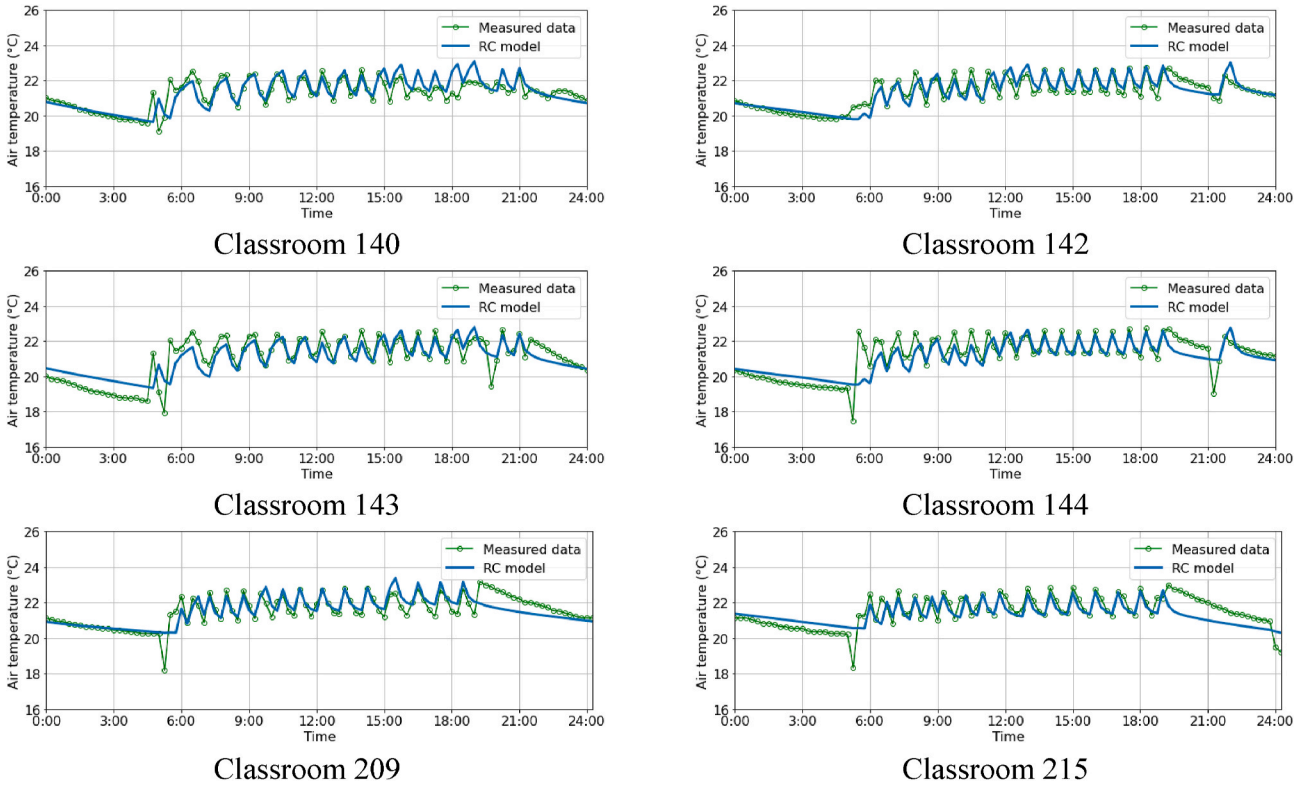


Fig. 6. Calibration of the thermal network model with measured data in different classrooms.

Table 2
Comparison between the grey-box models and the measured data across different classrooms.

Classrooms	First-order model		Second-order model		Third-order model		Fourth-order model	
	CV(RMSE) %	NMBE %	CV(RMSE) %	NMBE %	CV(RMSE)%	NMBE %	CV(RMSE) %	NMBE %
Classroom 140	24.82	15.28	17.25	14.12	15.88	11.16	8.71	4.24
Classroom 144	26.12	17.12	19.04	16.19	16.53	14.43	10.15	4.13
Classroom 215	33.11	18.15	20.12	16.13	17.18	15.55	5.43	3.86
Classroom 142	31.07	18.98	20.09	16.54	18.99	14.46	5.11	3.19
Classroom 143	35.32	21.12	22.65	19.98	20.06	15.13	7.33	4.86
Classroom 209	33.19	20.23	21.07	18.87	20.44	16.89	8.01	4.92

$$where\ J = \left(\sum_{i=1}^{N}\left(P^i \Delta t\right)\cdot Cost^i\right) + \left(\sum_{i=1}^{N}\left(\alpha \cdot d^i\right)\right)$$

Subject to $\underline{T} \le T_{in} \le \overline{T}$

$0 \le P \le P_{max}$

$$d^i = \begin{cases} \left(\underline{T}^i - T^i\right)\Delta t & T^i < \underline{T}^i \\ 0 & \underline{T}^i \le T^i \le \overline{T}^i \\ \left(T^i - \overline{T}^i\right)\Delta t & \overline{T}^i > T^i \end{cases} \quad (8)$$

where the control variable (P_i) is the heating power demand at the time step, Δt is the time step, *cost* is the price of electricity (¢/kWh), and N is the number of time steps across the prediction horizon PH. Thermal discomfort was considered as a soft constraint, penalizing it if the indoor temperature of a zone was lower than the setpoint, or if it was greater by more than 3 °C. α is a factor that weighs the cost of energy to the 'cost' of discomfort. It needs to be large enough that thermal comfort violations are penalized adequately. The space heating demand (P) is limited by the maximum capacity of the heating equipment (P_{max}). In this study, MPC performs optimization starting at midnight with a 24-h prediction horizon and timestep of 15 min. The rest of the constraints force the MPC formulation to use the building model and restrict the heating output to the installed capacity of each zone. The optimization problem was solved using the cvxpy library in Python [55].

3.5. Model predictive control implementation and energy flexibility quantification

The MPC optimization was performed offline each night using the most recent measured building operation data and one day-ahead weather forecasts. The resulting near-optimal temperature setpoint profiles were then transmitted via the BACnet communication protocol to the building management system and executed in real time over the following day. On average, each day's optimization problem was solved in less than 3 min on a standard laptop (Intel i7, 16 GB RAM), making it computationally efficient and well-suited for overnight batch processing. This field-based validation, bridging the gap between advanced control theory and practical building management applications. The prediction horizon of 24 h allows for a comprehensive forecast of future electricity prices and heating requirements. Fig. 7 presents the framework of the MPC methodology used in this study.

Energy flexibility in building net electricity demand is an essential part of the solution to address the electrical future grids requirements. Dynamic energy flexibility is defined as the capability of a building to

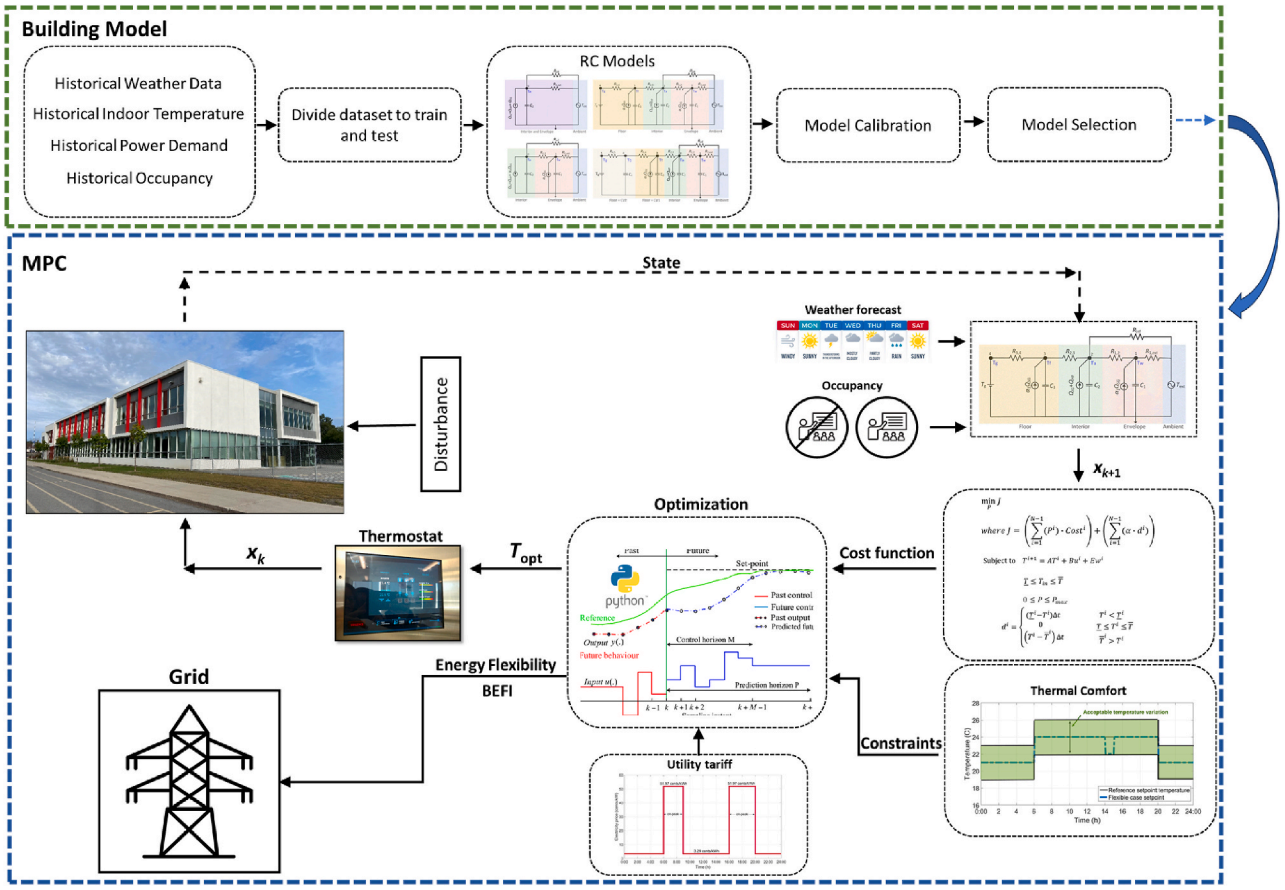


Fig. 7. MPC framework used in this study.

reduce or increase its electricity demand for a period of time needed for the grid. Energy flexibility has been calculated based on Equations (9) and (10) [56]. This equation calculates the building energy flexibility index (BEFI) under the implementation of the flexibility strategy and the reference as-usual profile. In this equation, the P_{ref} refers to the power demand (i.e., HVAC loads) under business-as-usual operation, which is the current operation of the school. The P_{flex} represents the HVAC power demand under the MPC controller, considered the flexible case, and the Dt is the duration of the flexibility event needed by the grid. We presented the details of this index in Ref. [56]. The calculation of the BEFI as a percentage compares the peak power under the flexible case and the reference as usual profile (equation (4)). The BEFI provides a measure of the potential reduction in peak power demand achieved through implementing the MPC controller. A higher BEFI indicates greater flexibility and potential for demand response. The calculation of the BEFI takes into account both the magnitude and duration of the flexibility event, providing a comprehensive assessment of its impact on power demand.

$$\overline{BEFI}(t, Dt) = \frac{\int_{t}^{t+Dt} P_{Ref}\,dt - \int_{t}^{t+Dt} P_{Flex}\,dt}{Dt} \tag{9}$$



Fig. 8. School building (right) and inside classroom with a smart control panel (left).

$$BEFI\% = \frac{\overline{BEFI}(t, Dt)}{P_{Ref}} \times 100 \quad (10)$$

4. Description of case study building

The demonstration site is an elementary school located in Quebec, Canada, and the MPC discussed in the later sections was deployed to the classrooms in the school, as indicated in Fig. 8. The school building consists of various classrooms equipped with water-to-air heat pumps. The school is located in Sainte-Marthe-sur-Lac, near Montreal, QC, Canada, makes it an ideal site for studying the impact of flexibility events on power demand in educational institutions in cold climate regions. This two-storey school is 80.7 m by 43.8 m and oriented 35° west of south. The first floor consists of 10 classrooms, offices, a kitchen, a daycare, a music room, a kindergarten, and a gym. The second floor has 13 classrooms and a library. It is an archetype of new school buildings in Québec, with a 60 % window-to-wall ratio of double-glass windows with an air gap. The classrooms are heated with local water-to-air HPs, and an individual thermostat in each classroom. Each classroom has a smart controller panel to adjust the temperature and lighting level and monitor CO_2 ppm, relative humidity, and energy consumption (Fig. 9).

The school also incorporates energy-efficient LED lighting and motion sensors to reduce energy consumption further. A detailed summary of the building's systems, including HVAC configurations, HVAC schedules, and utility tariffs, is provided in Table 3. Thermostat and power meter data were collected at 15-min intervals in CopperCube, an on-site trend log archiver, through the BACnet network.

The experiment is designed to evaluate the effectiveness of MPC in reducing peak load and energy consumption during on-peak hours. It also examines the influence of occupancy on MPC performance and energy flexibility by including both occupied and unoccupied periods, thereby assessing its impact across different operational scenarios. By conducting the experiment during cold winter days, we were able to simulate real-life conditions where electrical power demand is typically high. For this purpose, eight classrooms were investigated (as shown in Figs. 10 and 11), including four on the first floor and four classrooms on the second floor with different orientations (as much as possible). Four classrooms were considered the reference cases (Blue highlights), and six classrooms were considered the flexible cases (Green highlights).

The reference cases followed the conventional thermostat control strategy, while the flexible cases utilized MPC to optimize energy consumption. In this way, we could compare the results of the MPC implementation with those of the business-as-usual one. We aimed to capture a diverse range of thermal conditions by including classrooms

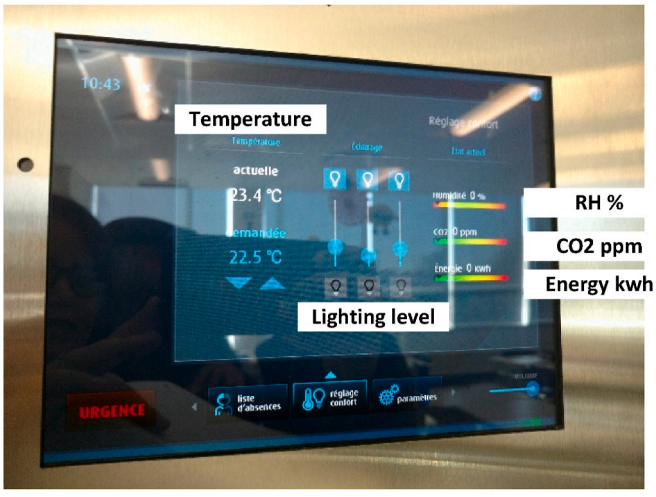


Fig. 9. Control panel for adjusting setpoint temperature and monitoring measured data in classrooms.

Table 3
Key features of the school building and utility tariff.

General Information	Site	Sainte-Marthe-sur-le-Lac, Québec, Canada
	In operation	since 2017
	Net floor area	2596 m^2/floor
	Number of floors	2
	Number of classrooms	23
	Number of students	576
	Window type	Double-glazed argon Low-e
Mechanical	Space heating/ cooling	Hydronic radiant floor heating system, Convective system
	Heating/cooling equipment	Ground source water-water HP and local water-air HPs
	Ventilation system	Centralized dual-core heat recovery system modulated based on CO_2
	Domestic Hot water source	Two electricity boilers with a capacity of 15 kW each
	Thermal Energy Storage	An electrical thermal storage unit with a capacity of 80 kW on the source side. (ThermElect)
Default Schedule	Occupied hours	From 06:00 to 18:00, 20–24 °C
	Unoccupied hours	From 18:00 to 6:00, 18–20 °C
Utility rate Flex M during winter	Demand charge	$15.16/kW
	Consumption during off-peak hours	3.29¢/kWh
	Consumption during on-peak hours: •Morning: 06:00 to 9:00 •Evening: 16:00 to 20:00	51.97¢/kWh

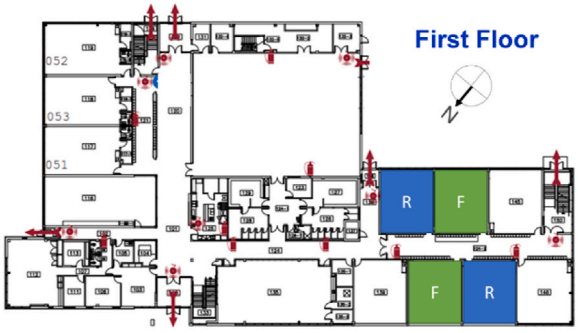


Fig. 10. Plan of the first floor of the school – green highlights: classrooms with MPC, Blue highlights: Reference classrooms.

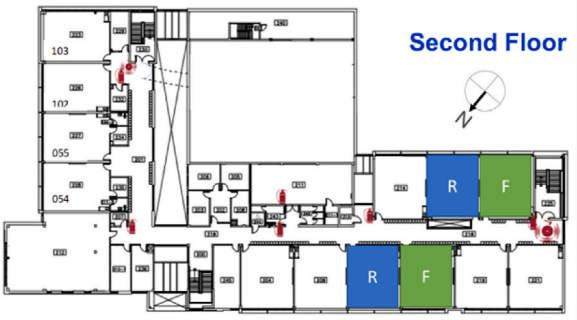


Fig. 11. Plan of the second floor of the school – green highlights: classrooms with MPC, Blue highlights: Reference classrooms.

Table 4
Characteristics of the selected classrooms and control strategies applied.

Room	Control strategy	Location	Orientation
Classroom 140	Reference	First Floor	Southeast
Classroom 142	MPC	First Floor	Northwest
Classroom 143	MPC	First Floor	Southeast
Classroom 144	Reference	First Floor	Northwest
Classroom 146	MPC	First Floor	Northwest
Classroom 209	MPC	Second Floor	Northwest
Classroom 215	Reference	Second Floor	Northwest
Classroom 216	Reference	Second Floor	Southeast

on both floors with varying orientations. Table 4 presents each classroom's control strategy, location, and installed heating capacity.

5. Results and discussion

Four control scenarios are investigated for cold winter days to evaluate the performance of different temperature control strategies in classrooms. The first scenario, Reference Scenario I, employed PI controllers in selected classrooms without considering occupancy. This setup represents a simplified control strategy where the system operates based on predefined setpoints and feedback loops, with no adjustments for occupant presence. The second scenario, Reference Scenario II, involved the use of PI controllers in classrooms while accounting for occupancy. This represents the business-as-usual (BAU) approach commonly implemented in conventional systems. In this setup, controllers adjust temperatures based on real-time data, such as indoor temperature and occupancy status, providing a more responsive solution compared to Reference Scenario I. However, as a baseline strategy, it lacks advanced optimization capabilities and relies on fixed rules for operation. The third scenario, Flexibility Scenario I, introduced MPC in classrooms without considering occupancy. Unlike PI controllers, MPC leverages real-time data, weather forecasts, and system models to predict future states and optimize temperature setpoints dynamically. This approach demonstrated improved energy flexibility by proactively adjusting heating operations based on external conditions. However, the lack of occupancy in this scenario may lead to energy inefficiencies during occupied periods. The fourth scenario, Flexibility Scenario II, further enhanced the application of MPC by integrating real-time and forecasted occupancy data. Including occupancy in the MPC strategy enables more precise and adaptive temperature control by aligning thermal conditions with actual room usage patterns. The inclusion of occupancy data resulted in improved energy savings, particularly during high-demand periods such as early morning hours, while simultaneously maintaining occupant comfort. By dynamically balancing energy efficiency and comfort, this approach demonstrated the potential of advanced control strategies to achieve optimal performance in building systems.

The summary of control scenarios is as follows.

a) Reference scenario I: PI controller in classrooms without occupancy
b) Reference scenario II: PI controller in classrooms with occupancy
c) Flexibility scenario I: MPC in classrooms without occupancy
d) Flexibility scenario II: MPC in classrooms with occupancy

5.1. Comparing model predictive control with proportional-integral controller

Four classrooms operated with PI controller and usual setpoint profiles were considered reference cases, while another four classrooms employed model predictive control as flexible cases. The current classroom operation, considered as the reference case, is based on a reactive controller for the local HPs. In this scenario, as shown in Figure 12a, the indoor temperature is around 22 °C during occupied hours (6:00 to 18:00), and the peak load occurs during the morning peak hours (6:00 to 9:00 a.m.) shown in Figure 12b. The energy consumption during high-price periods for the reference case was in the range of 2.6 kWh to 5.1 kWh, with an average of 3.5 kWh. These high-price periods can significantly impact energy costs for the schools. Also, during the reference scenario, the reactive controller started heating the zones as soon as the setpoints changed. The time it took for the zones to reach the setpoints resulted in some thermal discomfort. This thermal discomfort was particularly noticeable during the morning peak hours when the reactive controller struggled to heat the zones to the new setpoints quickly. This delay in reaching the desired temperature could decrease productivity and create discomfort for students and staffs. Therefore, it is crucial to find a solution that minimizes this thermal discomfort while still optimizing energy consumption during high-price periods. By implementing a more proactive approach that considers occupancy patterns, weather conditions, and utility tariffs, energy consumption during these high-price periods can be optimized. This would reduce energy costs and improve comfort levels for the occupants throughout the day.

5.2. Control scenarios without occupancy (reference and flexibility scenarios I)

As shown in Fig. 12a, the indoor temperature profile for the reference case and the flexible case without occupancy (school's holidays) reveals distinct differences in their operation. During the reference case, the indoor temperature remains nearly constant during occupied hours (6:00 to 18:00) to meet fixed setpoints. In contrast, the flexible scenario demonstrates a dynamic temperature control strategy, where indoor temperatures drop by approximately 2 °C during morning peak hours (6:00 to 9:00 a.m.) to reduce energy consumption. This temperature adjustment, facilitated by MPC, aligns with energy-saving goals while ensuring the space remains within its comfort boundaries. The CO_2 levels, shown in Fig. 12c, indicate that the indoor air quality remains within acceptable ranges, even with reduced heating operations, ensuring both sustainability and occupant well-being.

Figs. 13 and 14 illustrate the power demand comparison between the reference scenario I and flexible scenario I in different classrooms. In the

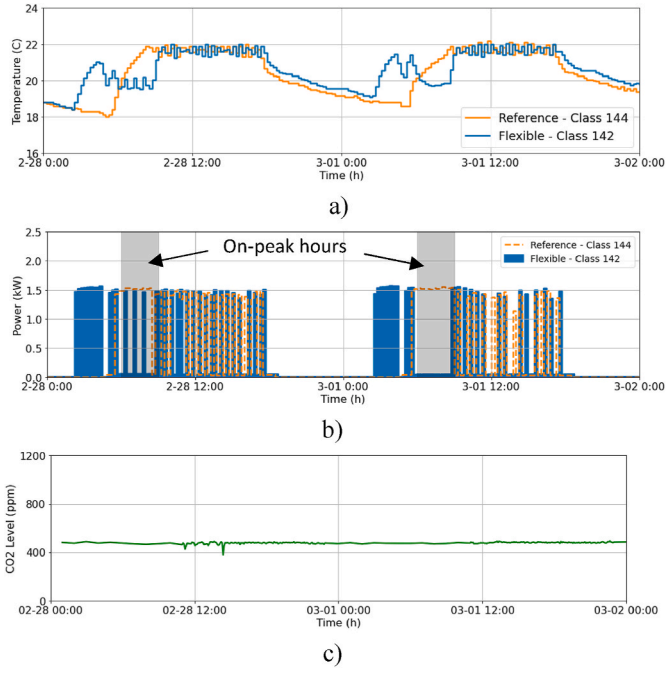


Fig. 12. Flexible scenario without occupancy. a) Indoor air temperature, b) power demand, and c) CO_2 level.

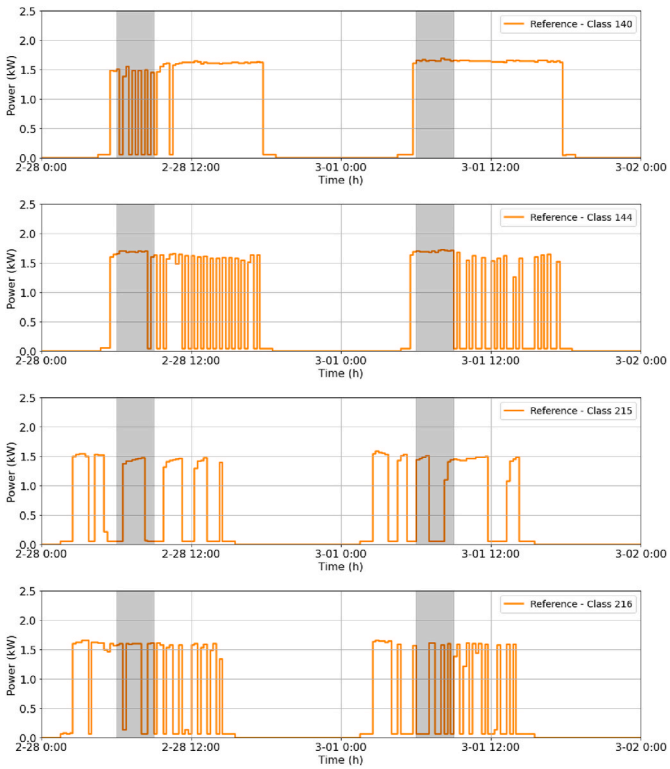


Fig. 13. Reference case, unoccupied classroom.

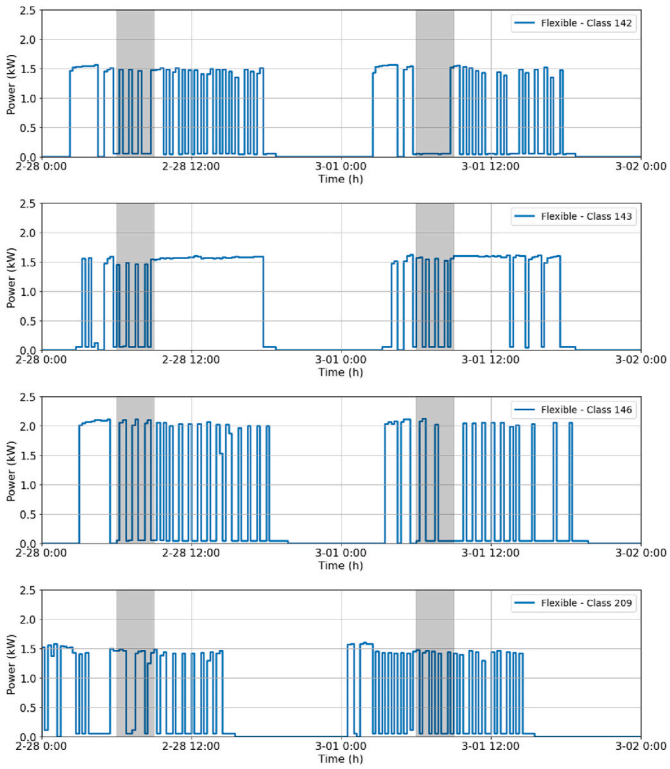


Fig. 14. Flexibility-enhancing scenario, unoccupied classroom.

reference case, the heating system reacts immediately to changes in setpoints, resulting in higher energy consumption during high-price periods. The energy consumption during high-price periods for the reference case was in the range of 2.3 kWh to 3.1 kWh, with an average of 2.5 kWh. These high-price periods can significantly impact energy costs for the schools. Also, during the reference scenario, the reactive controller started heating the zones as soon as the setpoints changed. The time it took for the zones to reach the setpoints resulted in some thermal discomfort. This thermal discomfort was particularly noticeable during the morning peak hours when the reactive controller struggled to quickly heat the zones to the new setpoints. This delay in reaching the desired temperature could decrease productivity and discomfort for students and staff. Therefore, it is crucial to find a solution that minimizes this thermal discomfort while still optimizing energy consumption during high-price periods. By implementing a more proactive approach that considers occupancy patterns, weather conditions, and utility tariffs, energy consumption during these high-price periods can be optimized. This would reduce energy costs and improve comfort levels for the occupants throughout the day. The flexible scenario employs a pre-heating strategy, where heating begins around 3:00 a.m. This strategy ensures that indoor temperatures reach desired levels before high-price periods commence, allowing the system to turn off heating entirely during the 6:00 to 9:00 a.m. peak hours. This pre-heating approach reduces the peak load by up to 65 %, significantly lowering strain on the grid and minimizing energy costs.

5.3. Control scenarios with occupancy (reference and flexibility scenarios II)

This section evaluates the impact of incorporating occupancy into the MPC strategy during field tests conducted under normal school operation. The classrooms were occupied by 20–25 students and a teacher, with real-time occupancy detection and CO_2 sensors providing critical inputs for dynamic control. The findings highlight the effectiveness of MPC in optimizing energy consumption while maintaining comfort and addressing user interventions. In this scenario, the MPC system was designed to minimize energy-related costs (e.g., energy consumption and peak demand) while ensuring indoor temperatures remained within a defined comfort band. The inclusion of occupancy data provided additional flexibility, allowing the system to adjust dynamically to varying internal heat gains caused by occupant activity. However, to ensure the credibility of the MPC system, override permissions for teachers were restricted through prior approval from the school board. This measure ensured the experiment operated under controlled conditions, preventing conflicts between user preferences and MPC actions that could otherwise compromise the study's outcomes.

In order to balance reducing peak demand with maintaining thermal comfort, the MPC framework explicitly considers indoor temperature constraints, ensuring that setpoint deviations remain within an acceptable thermal comfort band. The BEFI metric indirectly reflects this balance by measuring flexibility over predefined time periods, ensuring that reductions in energy use during peak hours do not cause discomfort. Additionally, the MPC strategy leverages pre-heating strategies, allowing the building to store thermal energy ahead of peak periods, thereby reducing the need for high energy demand during on-peak hours without compromising comfort. Discomfort is treated as a soft constraint in the MPC optimization problem, meaning that minor deviations ($\pm 0.5\ ^{\circ}C$) from the ideal setpoint may occur if they lead to significant energy savings, but large temperature violations are heavily penalized.

Fig. 15a demonstrates the indoor air temperature profile during the reference and flexibility scenario with occupancy. The system maintains a tighter control of indoor temperatures within the comfort band, leveraging real-time occupancy data to optimize temperature adjustments. As shown in Fig. 15b, by incorporating internal heat gains from occupants, the system avoids unnecessary heating, especially during peak occupancy periods, further contributing to energy savings. Fig. 15c illustrates the CO_2 levels during the flexibility scenario with occupancy. Despite reduced energy consumption and heating operations, CO_2 levels remained within acceptable limits, indicating that air quality was not

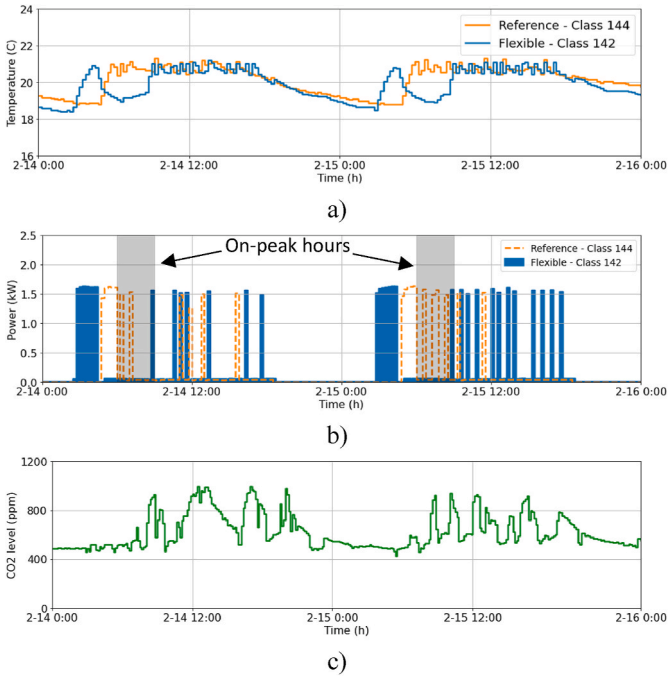


Fig. 15. Flexible scenario with occupancy. a) Indoor air temperature, b) power demand, and c) CO_2 level.

compromised. This highlights the system's ability to balance energy efficiency with indoor environmental quality, a critical factor in occupied spaces such as classrooms.

Figs. 16 and 17 present the power demand comparison between the reference scenario II and flexible scenario II. Compared to the unoccupied flexibility scenario (Flexibility Scenario I), the inclusion of occupancy data results in a 30 %–40 % reduction in power demand during

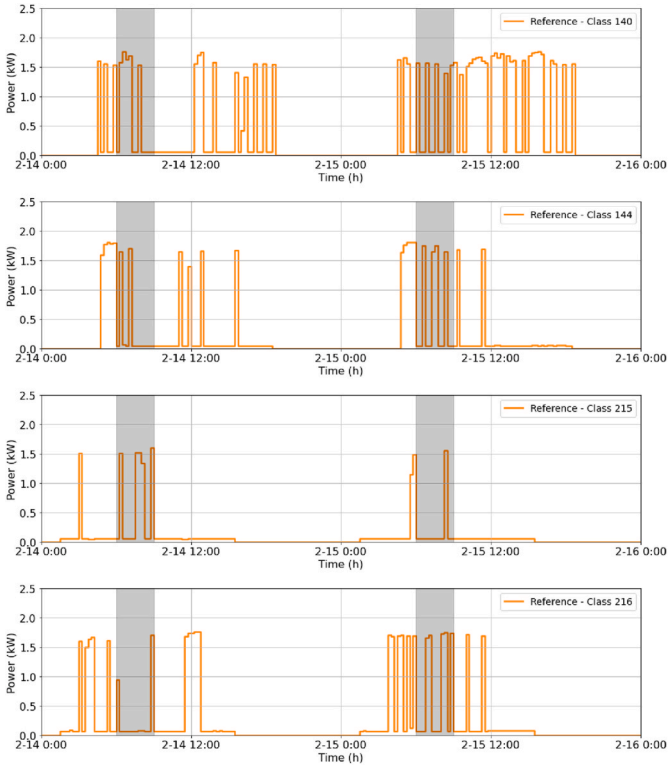


Fig. 16. Reference case, occupied classroom.

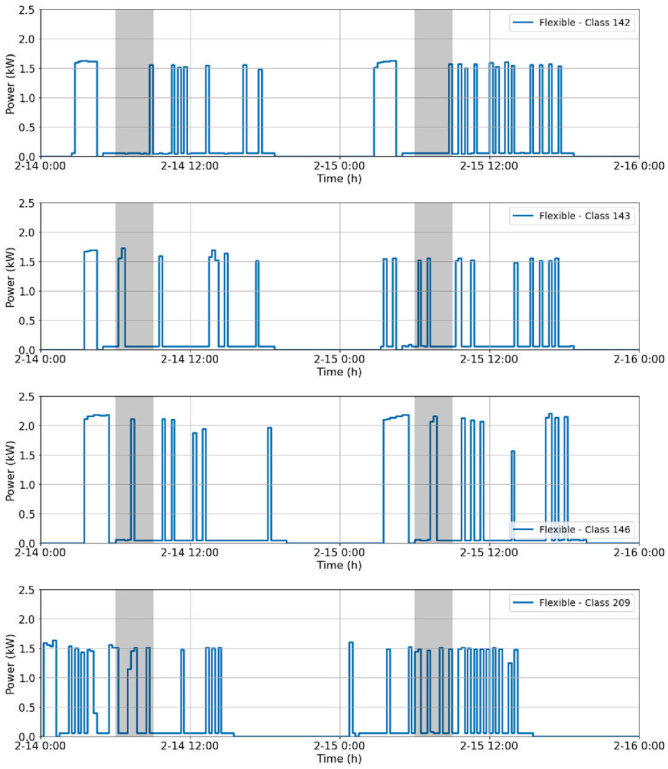


Fig. 17. Flexibility-enhancing scenario, occupied classroom.

occupied hours. Additionally, energy consumption in the flexibility scenario with occupancy is 45 % lower than in Flexibility Scenario I, emphasizing the importance of accounting for occupant presence in predictive control strategies. As shown in Fig. 16b, a sudden decrease in power demand is observed near 12:30p.m., corresponding to lunchtime when students temporarily leave the classrooms. Upon their return, internal heat gains from occupants significantly reduce the heating load required to maintain the comfort band. This dynamic adjustment highlights the advantages of integrating real-time occupancy data into MPC systems, allowing them to adapt effectively to changing conditions.

5.4. *Energy flexibility quantification*

Figs. 18 and 19 illustrate the total power demand of a classroom under two conditions—without occupancy and with occupancy—across both reference and flexible control scenarios. The comparison highlights the role of internal heat gains from occupants and the effectiveness of MPC in reducing power demand during on-peak hours. The results show that classrooms with occupancy exhibit significantly lower power demand during on-peak hours compared to unoccupied classrooms. Specifically, the total power demand in occupied classrooms during these periods is approximately 50 % lower than in unoccupied classrooms. In the absence of occupants, the flexible scenario using MPC reduces power demand by 36 % compared to the reference scenario. This reduction is achieved through proactive strategies such as pre-heating during off-peak hours and dynamic temperature adjustments based on weather

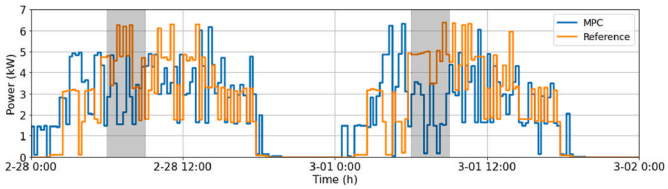


Fig. 18. Total power demand of the classrooms without occupancy.

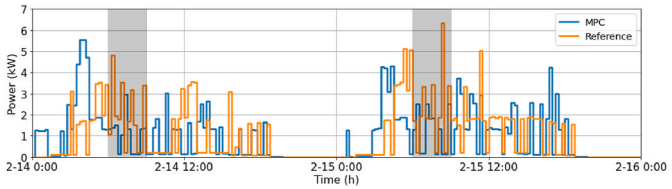


Fig. 19. Total power demand of the classrooms with occupancy.

forecasts and system predictions. When occupancy is considered, the reduction in power demand is even more pronounced, reaching 61 % under the flexible scenario. The integration of occupancy data enables the MPC to leverage internal heat gains to minimize energy consumption without compromising comfort.

Fig. 20 illustrates the power demand distribution across three scenarios: the reference case, MPC without occupancy, and MPC with occupancy, highlighting substantial energy savings and operational improvements achieved through the integration of advanced control strategies. In the unoccupied reference scenario, the median power demand is up to 90 % higher than in the flexibility scenario with occupancy, demonstrating the effectiveness of combining MPC with real-time occupancy data to minimize unnecessary heating and optimize energy usage. The flexibility scenarios show narrower and lower power demand distributions compared to the reference case, with the flexible scenario with occupancy achieving the lowest median and quartile ranges. This reflects smoother and more predictable power demand, reducing energy spikes and ensuring efficient operations. Energy consumption in the flexibility scenario without occupancy ranged from 10.1 kWh to 18.3 kWh, while incorporating occupancy reduced consumption further to 7.3 kWh to 12.6 kWh, leveraging internal heat gains to decrease heating requirements. On average, the MPC controller achieved a 49 % reduction in energy consumption compared to the reference case, underscoring its potential to enhance energy efficiency and operational consistency.

Figs. 21 and 22 illustrate the average BEFI during the experimental days for flexibility scenarios I (MPC without occupancy) and II (MPC with occupancy) compared to PI controller, highlighting the impact of internal heat gains and advanced control strategies on grid energy flexibility.

The BEFI is designed to quantify a building's ability to shift or reduce its electricity demand in response to grid requirements while ensuring that occupant comfort is maintained. Accounting for short-term load fluctuations, BEFI is calculated based on the difference between the reference (business-as-usual) power demand and the flexible case where MPC is implemented. Short-term fluctuations in loads—caused by internal heat gains, occupancy variations, or weather disturbances—are inherently captured in the BEFI formulation, as it is computed dynamically over the duration of flexibility events. The use of a 15-min time step in the analysis allows tracking rapid variations in power demand while ensuring that short-duration spikes or drops do not disproportionately affect the overall flexibility assessment.

In Flexibility Scenario I, the building provides up to 22 W/m^2 of energy flexibility during on-peak hours, corresponding to 44 % power

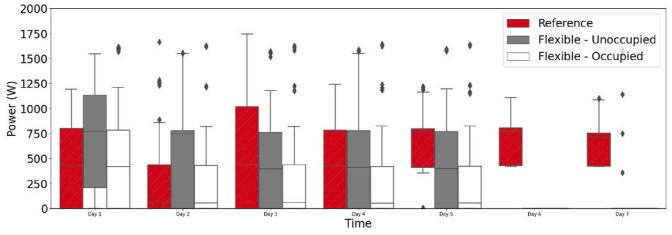


Fig. 20. Comparison of the power demand between Baseline, MPC without occupancy, and MPC with occupancy days in a classroom.

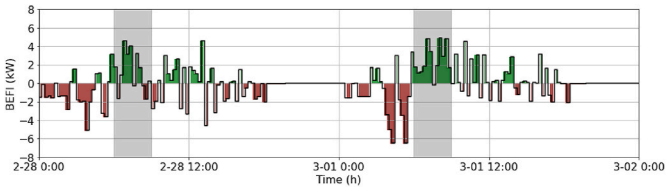


Fig. 21. Average BEFI for the flexible case without occupancy by using MPC. The shaded areas correspond to high-price periods.

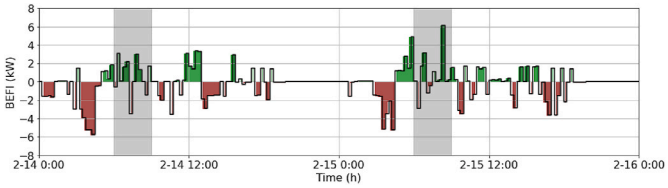


Fig. 22. Average BEFI for the flexible case with occupancy by using MPC. The shaded areas correspond to high-price periods.

demand reductions. However, in Flexibility Scenario II, the inclusion of occupancy data significantly enhances energy flexibility, with values increasing to 38 W/m^2 during on-peak hours, representing a 76 % reduction in power demand. The negative BEFI observed in the early morning reflects the energy demand for preheating before the morning peak. This effect is more pronounced in Flexibility Scenario I, where the MPC controller without occupancy preheats earlier and more intensively, leading to higher energy consumption and a larger negative BEFI. Conversely, in Flexibility Scenario II, the integration of occupancy data allows for more accurate heat demand prediction, resulting in optimized preheating schedules and reduced energy consumption. Including occupancy in MPC controllers improves energy flexibility by aligning space heating with actual usage. These findings underscore the importance of integrating occupancy data into predictive control systems to maximize energy efficiency and grid responsiveness.

Table 5 presents the comparison between the results of MPC and PI Controller and summarizes the quantification of energy flexibility across various classrooms and control scenarios, emphasizing the superior performance of MPC with occupancy prediction compared to PI controller.

By effectively managing school energy usage, the MPC can contribute to overall grid stability and reliability. Additionally, the ability to tap into the demand response resources of schools can help support the integration of renewable energy sources into the grid, further advancing sustainability goals.

5.5. *Challenges and future work*

This section outlines the challenges encountered during the experiment and the corresponding mitigation measures implemented to address them.

- During the experiment, there was a potential challenge of occupants overriding the pre-set temperature schedules, which could disrupt the control strategy and undermine the reliability of the data. To mitigate this, access to override the indoor temperature was restricted by obtaining formal permission from the school board. This ensured that the experiment was conducted under controlled conditions, allowing the MPC system to function as intended without interference. Such measures also maintained consistency across all scenarios and minimized deviations caused by individual user actions.
- Incomplete data and missing measurements may cause controller not to perform optimally. To mitigate these challenges in this study, the

Table 5
Energy Flexibility quantification in different classrooms and control scenarios.

Scenario	Room	Without Occupancy Energy Consumption during peak hours (kWh)	Average BEFI during peak hours (W)	BEFI% during peak hours	With Occupancy Energy Consumption during peak hours (kWh)	Average BEFI during peak hours (W)	BEFI% during peak hours
Reference with PI controller	Classroom 140	2.68	-	-	2.15	-	-
	Classroom 144	5.05	-	-	4.21	-	-
	Classroom 215	2.59	-	-	1.98	-	-
	Classroom 216	4.21	-	-	2.60	-	-
	Average	**3.46**	-	-	**2.73**	-	-
MPC	Classroom 142	1.96	268	26	0.54	536	75
	Classroom 143	2.34	536	54	0.97	810	77
	Classroom 146	2.39	77	7	0.66	330	66
	Classroom 209	3.68	125	58	1.89	177	27
	Average	**2.59**	**251**	**36**	**1.01**	**463**	**61**

following measures were implemented: In cases where data is missing, linear interpolation and backfilling techniques were applied to estimate missing values based on historical trends and surrounding data points. This approach ensures continuity in control operations without abrupt disruptions. Despite these mitigations, data reliability is recognized as a key factor in achieving optimal control performance. Future improvements could include even when sub-optimal data is available, the controller prioritizes constraints related to occupant comfort and operational safety, ensuring that extreme deviations are avoided. Implementing advanced machine learning-based fault detection or sensor redundancy strategies to further enhance robustness could be also considered.

- Flexibility measures, such as reducing heating during peak periods, posed a potential risk of discomfort for occupants if not implemented carefully. To ensure thermal comfort was maintained, continuous communication and collaboration were established with the school facility managers and school board. This proactive approach allowed any adjustments to the setpoint temperature to be made thoughtfully, prioritizing occupant well-being while achieving energy savings. Additionally, real-time monitoring of indoor environmental quality ensured that the comfort band was not violated, balancing energy efficiency with occupant satisfaction.
- Many school buildings rely on outdated infrastructure that may not support seamless integration with advanced control technologies. Retrofitting these systems to accommodate MPC can be technically challenging and costly.
- Schools involve diverse stakeholders, including utility companies, facility managers, school boards, teachers, parents' associations, and children. Resistance to adopting new technologies often arises due to unfamiliarity, perceived risks, or insufficient communication about the benefits of MPC. Engaging stakeholders through clear demonstrations of cost savings, comfort improvements, and environmental benefits is crucial.

By addressing these challenges with effective mitigation strategies, the study demonstrates the feasibility of integrating MPC systems in school buildings while ensuring energy efficiency, operational reliability, and occupant satisfaction. These insights also serve as a foundation for scaling and replicating similar systems in other educational facilities.

6. Conclusion

This paper presented the field implementation and assessment of an MPC framework in an all-electric school building with focusing on impact of occupancy on energy flexibility. Four scenarios have been investigated and compared: 1) Reference case with a reactive controller (PI controller) in unoccupied classrooms, 2) Reference case with a PI controller in occupied classrooms, 3) Flexible scenario using MPC in unoccupied classrooms, 4) Flexible scenario using MPC in occupied classrooms.

The MPC was implemented in eight classrooms, with six using the MPC and four as reference cases. The energy flexibility is quantified based on a building energy flexibility index (BEFI). The results revealed that the MPC system significantly outperformed the conventional PI controllers in terms of energy flexibility and efficiency. Specifically, the MPC improved energy flexibility by 36 % under unoccupied conditions and by 61 % when classrooms were occupied. Additionally, it reduced energy consumption by 25 % in unoccupied scenarios and by an impressive 63 % in occupied scenarios. Including occupancy in MPC controllers improves energy flexibility by aligning space heating with actual usage. The integration of occupancy data enables the MPC to leverage internal heat gains to minimize energy consumption without compromising comfort. This approach optimizes the use of internal heat gains and enables better timing for predictive control strategies. These findings highlight the effectiveness of integrating occupancy-based strategies into advanced control systems, as the increased internal heat gains during occupancy allowed the MPC to optimize preheating strategies and improve demand forecasting accuracy. Moreover, the MPC consistently maintained thermal comfort and indoor air quality, demonstrating its ability to balance energy performance with occupant needs. Moreover, this study underscores the critical importance of energy flexibility in cold climate contexts, where increased heating demand during early mornings overlaps with high electricity pricing and grid constraints. The proposed MPC framework is enabling predictive preheating and load shifting to alleviate stress on the electrical grid. These strategies are essential for optimizing energy use in all-electric buildings and aligning thermal comfort with utility demand profiles in regions with severe winter conditions.

This study demonstrates how MPC can serve as a key enabling technology for achieving grid-interactive, energy-efficient educational facilities, paving the way for broader implementation in other building types and climates. Nevertheless, limitations still remain, particularly in long-term predictions, due to unmeasured disturbances and theoretical model errors inherent to the MPC infrastructure. Future work should focus on these limitations and enhance the performance of the MPC system for schools. Also, future work will explore comparisons with other state-of-the-art control methods, such as reinforcement learning-

based approaches and hybrid optimization frameworks.

CRediT authorship contribution statement

Navid Morovat: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Andreas K. Athienitis:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization. **José Agustín Candanedo:** Writing – review & editing, Supervision, Resources, Conceptualization. **Hervé Frank Nouanegue:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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